Software Test Plan

Hostel Management System

Test documentation aligned with ISO/IEC/IEEE 29119-3:2021

Project	Hostel Management System
Document Version	1.0
Prepared for	Software Engineering Mini-Project

1. Introduction

1.1 Purpose

This Test Plan defines the testing scope, approach, responsibilities, environment, entry/exit criteria, and traceability for the Hostel Management System. Test cases are maintained in a companion document so that the plan remains consistent with the project deliverable structure.

1.2 Scope

Testing covers authentication, student records, room records, room allocation/deallocation, occupancy information, authorization, data integrity, performance, and usability requirements defined in the SRS.

1.3 Test Objectives

- Verify each in-scope functional requirement against its expected behavior.
- Validate selected measurable non-functional requirements and security controls.
- Verify that invalid and conflicting room allocations are rejected without corrupting data.
- Maintain traceability between SRS requirements and planned test cases.

1.4 References

The primary system baseline is the SRS. Test documentation is structured with reference to [2].

2. Test Items

Test item	Description
Authentication module	Login, authorization, session handling, and protected endpoints.
Student module	Create, read, search, and update student information.
Room module	Room creation, room capacity, occupancy display.
Allocation module	Allocation, conflict prevention, modification, deallocation, and transactional integrity.
Security controls	Password storage, access control, HTTPS deployment, input validation, and security event logging.

3. Features to be Tested

Feature	Related requirements	Priority
Authentication and access control	FR-01, FR-09, SEC-01, SEC-02	High
Student record management	FR-02, FR-03	High

Feature	Related requirements	Priority
Room management	FR-04	High
Room allocation	FR-05, FR-06	High
Allocation modification/deallocation	FR-07	High
Occupancy information	FR-08	Medium
Data integrity	NFR-03	High
Performance	NFR-01	Medium
Availability monitoring and reporting	NFR-02	Medium
Usability	NFR-04	Medium
Maintainability / architecture inspection	NFR-05	Medium
Security controls	SEC-03 to SEC-06	High

4. Features Not to be Tested

The following are outside the stated project scope and are therefore not part of this Test Plan: fee collection/payment processing, mess/food management, visitor management, attendance management, complaint management, and biometric access control. Their exclusion is a scope decision, not an assessment of implementation quality.

5. Test Approach

5.1 Security Validation

Security validation shall verify authentication and authorization boundaries, protected resource access, credential-storage behavior, input validation, transport protection in deployment, and security-event logging. Validation shall include both positive and negative tests and shall inspect that a security control is enforced at the server/API boundary rather than only in the user interface.

Security control	Validation method	Expected evidence
SEC-01 Authentication	Attempt protected access before and after valid login.	Unauthorized request rejected; authenticated request permitted.
SEC-02 Authorization	Use student credentials to invoke administrator-only operations.	Operation denied with an authorization response; no state change.
SEC-03 Password protection	Inspect configured credential storage and test authentication behavior.	No plaintext password stored; valid authentication remains possible.
SEC-04 HTTPS/TLS	Inspect deployment URL/configuration and reject insecure application access where deployment policy requires HTTPS.	Protected traffic uses HTTPS/TLS.
SEC-05 Input validation	Submit malformed/hostile identifiers and fields.	Input rejected safely; no unintended database behavior.
SEC-06 Security logging	Perform authentication and allocation security events and inspect logs.	Events contain actor, timestamp, event type, and outcome as defined by the implementation.

5.2 Test Levels and Types

Level/type	Purpose
Functional testing	Verify externally observable behavior for each FR.
Integration testing	Verify controller/service/database interactions, especially allocation transactions.

Level/type	Purpose
System testing	Verify end-to-end hostel-management workflows.
Security testing	Verify access control, input validation, credential protection, and security logging.
Performance testing	Measure NFR-01 under the specified workload target.
Usability testing	Measure the administrator workflow against NFR-04.
Availability verification	Review monitoring, outage detection, recovery records, and monthly availability reporting for NFR-02.
Maintainability verification	Inspect architecture/design documentation and modularity evidence for NFR-05.

5.3 Test Data

Test data shall include at least two student records, multiple rooms with different capacities, an available room, a full room, and a student with an existing active allocation. Test data shall be synthetic and shall not use real student personal information.

5.4 Test Environment

Item	Planned environment
Client	Modern desktop web browser.
Application	Hostel Management System build under test.
Database	Relational database configured with representative hostel data.
Network	Local or controlled network for functional tests; controlled HTTPS deployment for security validation.
Performance	Environment capable of generating up to 50 concurrent authenticated test sessions.

5.5 Entry Criteria

- Build is deployable in the test environment.
- Database schema and seed test data are available.
- Authentication and required test accounts are configured.
- Requirements baseline is approved for testing.

5.6 Exit Criteria

- All planned high-priority functional and security test cases have been executed.
- No open Severity-1 defect remains.
- All failed high-priority cases are either resolved and retested or formally accepted as known deviations.
- Traceability from SRS requirements to test cases is complete for in-scope requirements.

6. Pass / Fail Criteria

A test case passes only when the observed result satisfies the expected result and all applicable preconditions are met. A test case fails when the required behavior is absent, incorrect, insecure, or materially outside the measurable acceptance threshold stated in the requirement.

7. Suspension and Resumption Criteria

Testing may be suspended when the application cannot be started, the test database is unavailable, or a blocking defect prevents execution of the majority of planned cases. Testing resumes after the blocking condition is removed and the environment is revalidated.

8. Test Deliverables

- Approved Test Plan.
- Detailed test cases and traceability matrix.

- Execution results and defect records.
- Security-validation evidence.
- Final test summary for the project milestone.

9. Responsibilities

Role	Responsibility
Test lead / assigned student	Maintain the Test Plan, test cases, traceability, and execution evidence.
Developer / implementation team	Provide deployable builds, fix defects, and support diagnosis.
Reviewer / faculty	Review requirements, test adequacy, and milestone deliverables.

10. Schedule

Testing shall be conducted after the corresponding implementation increment is available and before the milestone submission. Exact execution dates are project-team scheduling decisions and are not specified by the requirements brief.

11. Risks and Mitigations

Risk	Potential effect	Mitigation
Unstable build	Test execution blocked or results invalid.	Use a tagged test build and perform smoke checks before execution.
Inconsistent test data	Incorrect occupancy or allocation results.	Reset database to a known synthetic baseline before allocation tests.
Environment mismatch	Performance/security results not reproducible.	Record environment configuration with test evidence.
Requirement change	Traceability becomes stale.	Review requirement IDs before each test-cycle update.

12. Requirements Traceability

SRS requirement	Test case(s)	Purpose
FR-01	TP-01	Authentication
FR-02	TP-02	Student records
FR-03	TP-03	Student search
FR-04	TP-04	Room records
FR-05	TP-05	Valid allocation
FR-06	TP-06, TP-07	Conflict and duplicate prevention
FR-07	TP-08	Modify/deallocate
FR-08	TP-09	Occupancy
FR-09	TP-10	Student self-view
NFR-01	TP-11	Performance

SRS requirement	Test case(s)	Purpose
NFR-02	NFR02□AVAILABILITY.md	Availability monitoring and monthly reporting
NFR-03	TP-12	Transaction integrity
NFR-04	TP-13	Usability
NFR-05	Architecture inspection	Maintainability / architecture review
SEC-01 to SEC-06	TP-10, TP-14	Security validation

References

[1] ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*, IEEE, 2018.

[2] ISO/IEC/IEEE 29119-3:2021, *Software and systems engineering — Software testing — Part 3: Test documentation*, IEEE, 2021.

[3] ISO/IEC/IEEE 42010:2022, *Software, systems and enterprise — Architecture description*, IEEE, 2022.

[4] Software Engineering Mini-Project Deliverables Part-1, course handout, September 2026.